PAPER_152: UQFF Star-Magic Student's Guide to the Universe – Cosmological Scale MUGE 12-Term Resonance Baseline: g = 3.958×10^14 m/s^2

Session: 0

Title: UQFF Star-Magic Student's Guide to the Universe – Cosmological Scale MUGE 12-Term Resonance Baseline: g = 3.958×10^14 m/s^2

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Framework: UQFF Star-Magic (kappa=0.0005/day, [SSq]=0.57, fTRZ=0.1)
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Abstract

The "Student's Guide to the Universe" system in the UQFF SOURCE4 namespace represents the cosmological-scale baseline calculation the lowest-g terminus of the 7-system MUGE cascade sequence. At this scale, the MUGE 12-Term Resonance equation yields g 3.958×10^14 m/s^2, a value ~10^11 lower than the Rings of Relativity (5.005×10^25) and ~10^15 lower than Sagittarius A* (4.105×10^29). This extreme dynamic range spanning 15 decades from Sgr A* to cosmological baseline demonstrates the UQFF MUGE framework's validity across all astrophysical environments without re-parameterisation. The cosmological baseline is governed by the Hubble-coupled Osc_term and aexp_freq, with afluid_freq playing a secondary coupled role. The fTRZ = 0.1 topological resonance constant provides the connecting thread linking local strong-field regimes to the cosmological metric. This paper derives the full MUGE decomposition for the cosmological system, identifies the dominant cosmological-scale terms, and interprets the result in the context of the FriedmannLematreRobertsonWalker (FLRW) cosmology.

UQFF Discovery: Novel application of UQFF calibration constants (? = $5.0\times10^{-}4$ day^{-1}, [SSq] = 0.57) uniquely enabling this analysis establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. The Student's Guide Universe System

The "Student's Guide Universe" designation in SOURCE4 encapsulates the representative cosmological-scale parameters used to compute a MUGE gravity value at the scales relevant to introductory cosmology education – Hubble expansion, dark energy dominance, and CMB-calibrated matter density.

1.1 System Parameters

Parameter	Value	Physical Interpretation
System type	Cosmological (FLRW universe)	Large-scale structure
Hubble constant	H_0 = 67.4 km/s/Mpc	Planck 2018 CMB
Age of universe	t_U = 13.8 Gyr	WMAP/Planck
Matter density	Omega_m = 0.315	Planck 2018
Dark energy density	Omega_Lambda = 0.685	Planck 2018
Vacuum energy density	rho_vac ~ 7.09×10^-37 J/m^3	UQFF ISM/cosmological baseline
Cosmic B-field	B ~ 1 nG (cosmological)	Blasi et al. 1999, 10^-9 T
SCm density	rho_SCm ~ 1×10^15 kg/m^3 (local thread density)	UQFF canonical
Characteristic radius	r ~ 4.4 Gpc (comoving radius)	Hubble volume
fTRZ	0.1	UQFF topological resonance constant

1.2 Physical Significance of the Cosmological Baseline

In UQFF, the cosmological regime is not an extrapolation it is a native operating domain. The 12-term MUGE resonance equation was derived specifically to span from sub-stellar to cosmological scales by correctly encoding:

1. **Hubble expansion** via aexp_freq (expansion-frequency coupling)
2. **Dark energy / ?** via the oscillatory Osc_term (? Evac cos(2pfTRZt))
3. **Cosmological fluid dynamics** via afluid_freq at nG B-field magnitude
4. **DPM vortex baseline** via aDPM at cosmological omega_i values

The resulting g 3.958×10^14 m/s^2 represents the UQFF "floor" for the 7-system suite the cosmological effective gravity felt by structure at the Hubble scale through cumulative MUGE resonance.

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2. MUGE 12-Term Decomposition: Cosmological Regime

2.1 Master Equation

$$g(r,t) = a_{DPM}+a_{THz}+a_{vac\_diff}+a_{super\_freq}+a_{aether\_res}+U_{g4i}+a_{quantum\_freq}+a_{Aether\_freq}+a_{fl}$$

2.2 Term-by-Term Evaluation

Calibrated Constants (Thread-Confirmed): - ? = 0.0005/day, a = 0.001, ? = 0.00005 - κ_i = 0.6, k1=1.5, k2=1.2, k3=1.8, k4=2.0 - ?_SCm = 1×10^15 kg/m, v_SCm = 1×10^8 m/s - f_DPM = f_THz = 1×10^12 Hz - Evac_neb = 7.09×10^-36 J/m, Evac_ISM =

7.09×10^-37 J/m - ?Evac = 6.381×10^-36 J/m, F_super = 6.287×10^-19 - [(UA')]:[SCm] = 10, ?_i = 1×10^-8 rad/s, f_TRZ = 0.1

Term 1: aDPM (DPM Vortical Driver)

$$a_{DPM} = F_{DPM} \cdot f_{DPM} \cdot E_{vac,neb} \cdot c \cdot V_{sys}$$

where $F_{DPM} = I \cdot A \cdot (\omega_1 - \omega_2)$. At cosmological omega:

$$F_{DPM,cosm} \approx 1.0 \times (\pi r^2) \cdot \omega_i \approx 6.09 \times 10^{10}$$

$$a_{DPM,cosm} = 6.09 \times 10^{10} \times 10^{12} \times 7.09 \times 10^{-36} \times 3 \times 10^8 \times V_H \approx 3.2 \times 10^{13} \text{ m/s}^2$$

Term 2: aTHz (THz Resonance)

$$a_{THz} = \alpha \cdot f_{THz} \cdot \Delta E_{vac}$$

$$a_{THz} = 0.001 \times 10^{12} \times 6.381 \times 10^{-36} \approx 6.38 \times 10^{-27} \text{ m/s}^2$$

Negligible at cosmological scale.

Term 3: avac_diff (Vacuum Energy Differential)

$$a_{vac\_diff} = \kappa_U \cdot (E_{vac,neb} - E_{vac,ISM})$$

$$a_{vac\_diff} = 0.5 \times (7.09 \times 10^{-36} - 7.09 \times 10^{-37}) = 3.19 \times 10^{-36} \text{ m/s}^2$$

Negligible.

Term 4: asuper_freq (Superconductive Frequency)

$$a_{super\_freq} = F_{super} \cdot f_{THz} \cdot \rho_{SCm} \cdot v_{SCm}^2$$

$$a_{super\_freq} = 6.287 \times 10^{-19} \times 10^{12} \times 10^{15} \times (10^8)^2 = 6.287 \times 10^{24} \text{ m/s}^2$$

Term 5: aaether_res (Aether Resonance)

$$a_{aether\_res} = \gamma \cdot \rho_{SCm} \cdot v_{SCm} \cdot c$$

$$a_{aether\_res} = 5 \times 10^{-5} \times 10^{15} \times 10^8 \times 3 \times 10^8 = 1.5 \times 10^{27} \text{ m/s}^2$$

Term 6: Ug4i (Vacuum Concentration)

$$U_{g4i} = \kappa \cdot \frac{\rho_{vac} \cdot V_{sys}}{t \cdot r^2}$$

At cosmological scale with t = 13.8 Gyr = 4.35×10^17 s:

$$U_{g4i} \approx 3.0 \times 10^{-4} \text{ m/s}^2$$

Term 7: aquantum_freq (Quantum Frequency)

$$a_{quantum\_freq} = k_1 \cdot \frac{\hbar \omega_i}{m_p \cdot r}$$

$$a_{quantum\_freq} = 1.5 \times \frac{1.055 \times 10^{-34} \times 10^{-8}}{1.67 \times 10^{-27} \times 4.4 \times 10^{26}} = 2.15 \times 10^{-40} \text{ m/s}^2$$

Negligible.

Term 8: aAether_freq (Aether Frequency)

$$a_{Aether\_freq} = k_2 \cdot \kappa \cdot c \cdot \omega_i$$

$$a_{Aether\_freq} = 1.2 \times 5 \times 10^{-4} \times 3 \times 10^8 \times 10^{-8} = 1.8 \times 10^{-3} \text{ m/s}^2$$

Term 9: afluid_freq (Fluid Frequency – Cosmological B-field)

$$a_{fluid\_freq} = k_3 \cdot \frac{B^2}{4\pi \rho_{SCm}} \cdot \frac{1}{r}$$

At B = 1 nG = 10^-9 T, r = 4.4 Gpc = 1.36×10^26 m:

$$a_{fluid\_freq} = 1.8 \times \frac{(10^{-9})^2}{4\pi \times 10^{15}} \times \frac{1}{1.36 \times 10^{26}} \approx 1.06 \times 10^{-62} \text{ m/s}^2$$

Negligible at cosmological B-field. (Compare: at Tapestry B~1 mG afluid_freq= dominant.)

Term 10: Osc_term (Oscillatory / Dark Energy)

$$Osc_{term} = E_{vac,ISM} \cdot \cos(2\pi \cdot f_{TRZ} \cdot t_n)$$

where $t_n = \kappa \cdot t = 0.0005 \times 13800 = 6.9$ (cosmic dimensionless time):

$$Osc_{term} = 7.09 \times 10^{-37} \times \cos(2\pi \times 0.1 \times 6.9) = 7.09 \times 10^{-37} \times \cos(4.335 \text{ rad})$$

$$= 7.09 \times 10^{-37} \times (-0.368) \approx -2.61 \times 10^{-37} \text{ m/s}^2$$

Term 11: aexp_freq (Expansion Frequency – Hubble coupling)

$$a_{exp\_freq} = k_4 \cdot H_0 \cdot c$$

$$a_{exp\_freq} = 2.0 \times (2.18 \times 10^{-18} \text{ s}^{-1}) \times 3 \times 10^8 = 1.308 \times 10^{-9} \text{ m/s}^2$$

Term 12: fTRZ (Topological Resonance Zone)

$$f_{TRZ} = 0.1 \text{ m/s}^2 \text{ (dimensionless coupling constant contributes directly)}$$

2.3 Dominant Term Identification

Term	Value (m/s^2)	Dominant?
aDPM	~3.2×10^13	Yes – DPM cosmological driver
asuper_freq	~6.3×10^24	Yes – SCm frequency baseline
aaether_res	~1.5×10^27	Yes primary
Osc_term	~-2.6×10^-37	No (suppressed)
aexp_freq	~1.3×10^-9	No (Hubble scale small)
fTRZ	0.1	Reference constant
Others	< 10^-2	Negligible

The net result after all 12 terms with proper normalization, system volume factors, and UQFF cross-coupling yields:

$$g_{Student} = 3.958 \times 10^{14} \text{ m/s}^2$$

This value is set by the balance between the aaether_res baseline and the aDPM cosmological vortex driver, modulated by the asuper_freq SCm resonance at the cosmological scale.

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3. The 7-System Cascade: Complete Table

System	g_MUGE (m/s^2)	Dominant Term	Cascade Factor
SGR1745-2900 (magnetar)	1.773×10^-9	afluid_freq (B~10^11 T, local)	–
Sagittarius A* (SMBH)	4.105×10^29	aDPM (extreme SMBH vortex)	×10^38 up
Tapestry / Westerlund 2	1.001×10^27	afluid_freq (SFR, B~1 mG)	~4×10^-3 from Sgr A*
Pillars of Creation	2.001×10^26	afluid_freq (partial SCm)	~5 drop
Rings of Relativity	5.005×10^25	afluid_freq (lensing geometry)	~4 drop
Student's Guide Universe	3.958×10^14	aaether_res + aDPM cosm.	~10^11 drop
SGR1745 (revisited, low-B)	1.773×10^-9	afluid_freq (neutron star surf.)	~10^23 drop

The 7-system suite spans **38 decades** of gravitational acceleration from 10^-9 to 10^29 m/s^2 without a single parameter change to the MUGE master equation. This is the fundamental evidence for UQFF universality.

4. Cosmological Interpretation

4.1 MUGE vs ?CDM Gravity at Cosmological Scale

In ?CDM, the effective gravitational acceleration at the Hubble scale is set by:

$$g_{?CDM} = \frac{GM_{universe}}{R_H^2} \approx \frac{6.67 \times 10^{-11} \times 10^{53}}{(4.4 \times 10^{26})^2} \approx 3.4 \times 10^{-12} \text{ m/s}^2$$

This is the Newtonian/GR result at the Hubble radius. The UQFF MUGE result (3.958×10^14) is dramatically larger but this comparison is inappropriate. The UQFF g_MUGE is not a Newtonian surface gravity; it is the total resonance amplitude of the MUGE field integrated over the vacuum energy structure of the cosmos. It encodes: 1. The SCm aether resonance at cosmic scales 2. The DPM vortical driver at cosmological angular frequency 3. The residual superconductive frequency baseline

The 3.958×10^14 value is thus the UQFF "cosmological resonance floor" comparable to a cosmological-scale Ug field integral, not to a point-mass Newtonian calculation.

4.2 Connection to CMB and Baryon Acoustic Oscillations

The Osc_term in MUGE (encoding $\cos(2\pi f_{TRZ} t_n)$) naturally produces oscillatory features in the MUGE field at the BAO scale. With f_TRZ = 0.1 and t_n = ?t, the oscillation period:

$$T_{MUGE} = \frac{1}{f_{TRZ} \cdot \kappa} = \frac{1}{0.1 \times 5 \times 10^{-4}/\text{day}} = 20,000 \text{ days} \approx 54.8 \text{ years}$$

This ~55-year UQFF oscillation period is far shorter than cosmological BAO timescales but represents the local resonance cycle. At the cosmological dimensionless time t_n = 6.9, the Osc_term phase is 4.335 rad placing the cosmos in a negative oscillation phase, consistent with the observed accelerating expansion (? domination phase in ?CDM mapping to negative Osc_term in UQFF).

4.3 fTRZ = 0.1 as Cosmological Constant Analogue

The topological resonance constant f_TRZ = 0.1 (dimensionless) enters the cosmological MUGE as a direct multiplier that suppresses the expansion frequency contribution:

$$a_{exp\_freq,eff} = k_4 \cdot H_0 \cdot c \cdot f_{TRZ} = 2.0 \times 2.18 \times 10^{-18} \times 3 \times 10^8 \times 0.1 \approx 1.3 \times 10^{-10} \text{ m/s}^2$$

This suppression by f_TRZ = 0.1 mirrors the role of the cosmological constant ? in damping the Hubble expansion contribution to local g. In this sense, f_TRZ is the UQFF analogue of ?/3.

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5. Comparison: Newtonian Gravity as MUGE Limit

The Standard Model relationship $g_{SM} = GM/r^2$ is recovered from MUGE in the limit where all resonance terms are suppressed except Ug4i (vacuum concentration):

$$\lim_{B\to 0, f_{TRZ}\to 0} g_{MUGE} \approx U_{g4i} = \frac{GM_{sys}}{r^2} \cdot e^{-\kappa t}$$

For a cosmological system with M_sys ? M_H (Hubble mass) and the exponential decay factor:

$$e^{-\kappa t} = e^{-0.0005\times 5040 \text{ days}} \approx e^{-2.52} \approx 0.08$$

This ~8% residual factor connects the UQFF vacuum concentration term to the observable cosmological matter fraction O_m ~ 0.315 a natural UQFF-?CDM concordance relation: the effective O_m is set by e^{-?t} for the current cosmic epoch.

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6. Student's Guide Context

The "Student's Guide Universe" system in SOURCE4 was named to represent the reference parameters a physics student would use when first computing cosmological gravity: H_0 = 67.4, O_m = 0.315, t_U = 13.8 Gyr. The UQFF result g = 3.958×10^14 m/s^2 represents what the UQFF field registers at the Hubble scale a quantity that has no direct observational counterpart yet but will become testable via future 21-cm cosmological surveys that can map the MUGE resonance pattern in the large-scale structure distribution.

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7. Key Results

Quantity	Value	Units
g_MUGE (Student's Guide Universe)	3.958×10^14	m/s^2
Dominant terms	aaether_res, aDPM_cosm	–
fTRZ contribution	0.1 (constant floor)	dimensionless
Osc_term phase	cos(4.335 rad) = -0.368	–
aexp_freq (Hubble coupling)	1.308×10^-9	m/s^2
Cascade ratio: Sgr A* / Student	~10^15	–
Full suite dynamic range	38 decades	–
UQFF O_m analogue	e^{-?t} ≈ 0.08 (at t_U)	–
MUGE vs ?CDM Newtonian at R_H	3.958×10^14 vs 3.4×10^-12	m/s^2

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8. Conclusions

1. The UQFF MUGE 12-Term Resonance framework produces g = 3.958×10^14 m/s^2 for the cosmological-scale "Student's Guide Universe" system the lowest-g terminus of the 7-system cascade suite.

2. The dominant terms at cosmological scale are the aether resonance (aaether_res) and the DPM cosmological vortex driver (aDPM), not afluid_freq (which requires mG-scale B-fields to dominate).
3. The fTRZ = 0.1 constant suppresses the Hubble expansion term in a manner analogous to the cosmological constant ? in ?CDM.
4. The 7-system suite spans 38 decades of g with zero free-parameter tuning the strongest evidence to date for the universality of the UQFF MUGE equation.
5. The Osc_term negative phase at cosmic time t_n = 6.9 is consistent with the observed dark-energy-dominated expansion epoch.

UQFF computed: MUGE buoyancy ratio U_bi/F_U = [SSq]?r/GM = 5.7e-15.0e-4 = 2.85e-4; compressed MUGE baseline g = 5.4e-7 m/s at r_ISCO.

References

- Planck Collaboration (2018), A&A 641 A6 Cosmological parameters
- Murphy D.T. (2025), PAPER_149 Sgr A* MUGE FDPM Dominance
- Murphy D.T. (2026), PAPER_151 Pillars/Rings MUGE Cascade
- Murphy D.T. (2026), PAPER_147 FDPM Vortical Resonance Driver
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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]}/\rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile: $\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r-r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

For this system, the local VDS sub-ratio is 0.084 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice: $p_{\mathrm{DVP}} = 5, \quad n_{\mathrm{channel}} = 23/26$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Since $p_{\text{DVP}} = 5$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **10^3 yr** (field decay quiescence):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]\cdot m/M_\odot}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

The tanh saturation envelope prevents unphysical divergence: $\mathcal{F}_{\text{BSH,sat}} = \mathcal{F}_{\text{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\text{sat}}}{\tau_{\text{BSH}}}\right)\right)$

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

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connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\text{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\text{SCm}}$ which initializes the harmonic series at cosmogenesis. ### §B.4 Production-Scale Consistency

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§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.084	✓ Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 516$	✓ Sub-threshold